

Mission-Adaptive Reconfigurable Battery Management System for Autonomous Rovers

B.Tech Capstone Project — Proposal for Approval

Project Domain	Embedded Systems / Power Electronics
Platform	Rover / Unmanned Ground Vehicle
Core Innovation	SOC and SOA driven real-time battery topology reconfiguration
Battery Architecture	6-cell Li-Ion with three-switch reconfigurable topology

1. Introduction

Autonomous vehicles — rovers, drones, and underwater vehicles — depend critically on battery systems for mission success. The Battery Management System (BMS) is responsible for monitoring battery health, managing charge/discharge, and protecting cells from unsafe conditions. However, traditional BMS architectures treat the battery pack as a fixed electrical configuration and function purely as passive watchdogs — they monitor and disconnect, but cannot adapt.

Recent research in reconfigurable battery systems (RBS) has shown that dynamically switching cell connections between series, parallel, and bypass configurations can dramatically improve energy utilization, enable fault tolerance, and extend pack lifetime. However, this technology has been explored primarily at a theoretical level for large-scale energy storage. No study has implemented a mission-adaptive reconfigurable BMS on a small autonomous vehicle.

This project bridges that gap by designing and building a reconfigurable BMS for an autonomous rover that uses State of Charge (SOC) and Safe Operating Area (SOA) as real-time inputs to dynamically alter battery topology — enabling the rover to continue operating even when individual cells fail, degrade, or become imbalanced.

2. Problems in Traditional BMS

2.1 Stranded Energy (Weakest Cell Effect)

In a fixed series pack, the same current flows through every cell (Kirchhoff's current law). When any single cell reaches its undervoltage cutoff, the entire pack shuts down — even if the remaining cells still hold 30–50% charge. This energy is completely inaccessible. After 100+ cycles, cell-to-cell SOC drift of 5–15% is common due to manufacturing variance and uneven thermal exposure. Even 10% imbalance means 10% of the pack's energy is wasted every cycle.

2.2 Binary Fault Response

When a traditional BMS detects any fault — overvoltage, undervoltage, overcurrent, or overtemperature — its only option is to disconnect the entire pack. For a rover mid-mission, this means total power loss. There is no graceful degradation and no ability to isolate a single problematic cell while continuing on the remaining healthy cells.

2.3 No Load Adaptation

A rover climbing an incline needs high current; cruising on flat ground needs less. A traditional BMS provides the same fixed voltage-current characteristic regardless of demand. Engineers must oversize the pack for worst-case scenarios, adding unnecessary weight for the 90% of the mission that does not require peak power.

2.4 Slow, Wasteful Cell Balancing

Passive balancing bleeds excess charge through resistors as waste heat. Active balancing requires additional inductors, capacitors, or transformers — adding cost, board space, and failure points. Both operate at milliamp-level currents against amp-level loads, making them fundamentally too slow to address imbalances during active operation.

2.5 Underutilized Intelligence

Traditional BMS systems estimate SOC and SOH but use them only for display purposes. These valuable state estimates do not feed into any decision-making loop that changes how the pack operates. The data exists but is never acted upon.

3. Proposed Solution

We propose a BMS that replaces the fixed battery topology with a dynamically switchable one. Each of the 6 individual Li-ion cells (21700 format) is connected through three N-channel MOSFETs — a series switch, parallel switch, and bypass switch — allowing the BMS to change cell connections in real time, under load, during a live mission. An STM32 microcontroller runs the reconfiguration algorithm and handles all monitoring and protection.

3.1 Cell Connection States

State	Series SW	Parallel SW	Bypass SW	Function
Series	CLOSED	OPEN	OPEN	Cell adds voltage to the string
Parallel	OPEN	CLOSED	OPEN	Cell shares current with adjacent cell
Bypass	OPEN	OPEN	CLOSED	Cell isolated, current bypasses it
Off	OPEN	OPEN	OPEN	Cell fully disconnected

Table 1: MOSFET switch states for each connection configuration

3.2 Nominal Configuration

The rover's electronics expect a stable ~11.1V supply. The nominal configuration is 3S2P — two parallel strings of three series cells. The BMS maintains this target voltage while dynamically managing which cells participate, how current is distributed, and which cells are bypassed or rested.

3.3 SOC-Driven Reconfiguration

Each cell's SOC is estimated using an Extended Kalman Filter. When a cell's SOC drops below a threshold (e.g., 20%) while others remain above 35%, the low-SOC cell is bypassed and remaining cells are reconfigured to maintain output voltage. This prevents the weakest cell from forcing a full shutdown and extracts usable energy from every cell individually.

3.4 SOA-Driven Reconfiguration

The Safe Operating Area is defined by voltage limits (2.5–4.2V), current limits (cell rating), and temperature limits (0–60°C). When any cell approaches or crosses an SOA boundary, the BMS immediately reconfigures — bypassing an overheating cell, redistributing current from an overstressed cell, or isolating a cell with voltage anomalies. The pack stays alive; only the problematic cell is removed from the active circuit.

3.5 Wear Leveling

The system tracks cumulative energy delivered by each cell. Cells that have worked more are periodically bypassed while fresher cells take over. This equalizes aging across all cells without wasting energy through balancing resistors — balancing happens naturally through workload management using the same MOSFETs already present for reconfiguration.

3.6 Protection Layer

All standard BMS protections are retained: per-cell overvoltage/undervoltage, pack-level overcurrent, short circuit detection, per-cell temperature monitoring, and pre-charge ramping. The critical difference is that these protections trigger reconfiguration as a first response. Full pack shutdown is the last resort, not the only option.

4. Reconfiguration Decision Triggers

The system evaluates four trigger categories every 100ms. Safety-critical SOA triggers take priority over efficiency-driven SOC triggers.

Trigger	Condition	BMS Action
High current detected	Per-string current exceeds 80% of cell rating	Add cells to parallel paths; redistribute current across more strings
Cell overheating	Cell temperature > 50°C or rising > 2°C/min	Bypass overheating cell; reconfigure remaining cells; cell cools passively
Voltage sag under load	Cell voltage drops below 3.0V while SOC > 20%	Cell has high internal resistance; bypass and flag for inspection
SOC imbalance	Difference between highest and lowest SOC > 15%	Bypass lowest-SOC cell; higher-SOC cells serve load; reintegrate when equalized
Cell SOC depleted	Any cell SOC drops below 20% threshold	Bypass depleted cell; continue mission on remaining cells
Load normalized	Total current returns below normal levels	Restore nominal 3S2P; rest previously overworked cells

Table 2: Reconfiguration trigger conditions and responses

Switching Transition Safety

Reconfiguration under load follows a safe sequence: motor PWM is briefly reduced (2–3ms), switches are opened on the outgoing configuration (break-before-make), new switches are closed, and PWM ramps back up. Total transition window is 5–10ms — imperceptible to the rover's drive system. A cooldown timer (minimum 500ms between topology changes) prevents oscillation.

5. Operational Scenarios and Edge Cases

5.1 Test Scenarios

#	Scenario	Traditional BMS	Our BMS	Expected Gain
1	Normal cruise (flat terrain)	Fixed config	Nominal 3S2P, all 6 cells active	Baseline
2	Steep incline (current spike)	No adaptation; cells overstressed	Parallel reconfiguration; current redistributed	Lower per-cell stress and heat
3	One cell overheats	Full pack shutdown	Bypass hot cell; continue on 5 cells	Mission continues at ~83% capacity
4	SOC imbalance (one cell low)	Full shutdown at weakest cell	Bypass low cell; extract remaining energy from others	15–30% more runtime
5	Cell fault (resistance spike)	Full pack shutdown	Bypass and isolate; continue mission	Fault tolerance; no mission failure
6	Extended operation	Uneven aging; no action	Rotate active/resting cells for wear leveling	Uniform degradation; longer pack life
7	Charging	Fixed series charging	All cells parallel; fast balanced charging	Faster charge; auto equalization

Table 3: Test scenario comparison matrix

5.2 Edge Cases

Edge Case	System Behavior
Two cells fault simultaneously	Both bypassed; controlled shutdown if remaining cells cannot maintain minimum voltage
All cells below 20% SOC	Low-power mode; reduce motor ceiling; alert vehicle controller to begin return-to-base
MOSFET stuck open	Detected via voltage sensing; cell excluded from all future configurations permanently
MOSFET stuck closed	Detected via unexpected current path; adjacent switches opened to isolate; emergency shutdown if isolation fails
Rapid load transients	Cooldown timer (500ms minimum) prevents oscillation between configurations
Ambient temp below 0°C	Under-temperature protection; charging disabled; discharge at reduced current only

Table 4: Edge case handling

5.3 Demo Setup

For the demo, the rover will be tested on flat surface and a ramp: (1) flat cruise in nominal config, (2) ramp climb showing current redistribution, (3) simulated cell fault via heat gun to trigger thermal bypass, (4) pre-imbalanced SOC test showing extended runtime. A laptop via CAN/UART displays real-time per-cell SOC, temperature, current, and active topology.

6. Advantages Over Traditional BMS

Metric	Traditional BMS	Our BMS	Improvement
Usable energy extraction	70–85% (weakest cell limits)	90–98% (per-cell extraction)	15–30% more runtime
Fault response	Full shutdown (0% output)	Bypass faulty cell (83% output, 5/6 cells)	Mission continuity
Cell balancing	Passive resistors (wastes energy)	Topology-based rotation (zero energy waste)	No balancing loss
Balancing hardware	External resistors, capacitors, inductors	Same MOSFETs used for reconfiguration	No extra components
Pack lifetime	Weakest cell degrades fastest (positive feedback)	Wear leveling equalizes (negative feedback)	20–40% longer pack life
Load adaptation	Fixed configuration	Dynamic parallel for high current demand	Reduced cell stress
Degraded operation	Failed pack = replacement	Operates on 4–5 cells at reduced capacity	Extends usable pack life

Table 5: Quantitative comparison

Energy Return Analysis

A 6-cell pack ($3.7\text{V} \times 5\text{Ah}$ each) holds 111 Wh. With 10% imbalance, a traditional pack strands 11–17 Wh. Our BMS overhead is ~ 0.5 Wh/hour. Net recovery: ~ 15 Wh — a 25x return on BMS energy cost.

7. Constraints and Limitations

Constraint	Impact	Mitigation
Scalability	Per-cell switching (3 MOSFETs/cell) does not scale to 1000+ cell packs	Hierarchical approach: cell-level in modules, module-level reconfiguration for large packs. Our project validates the concept at cell level.
MOSFET conduction losses	2–3 MOSFETs in current path add $\sim 300\text{mW}$ I^2R loss at 5A	Select ultra-low $R_{DS(on)}$ MOSFETs; loss is $<1\%$ of pack output power
Switching transients	5–10ms disruption during topology change	PWM ramp-down before switching; break-before-make sequence; 500ms cooldown timer
Added complexity	18 MOSFETs + gate drivers introduce new failure modes (stuck open/closed)	MOSFET health monitoring after each switch operation; safe fallback states
Algorithm validation	Incorrect switching could cause short circuits or current spikes	Rule-based deterministic logic (not ML); validated in MATLAB/Simulink before hardware deployment
PCB thermal management	18 MOSFETs on one board generate localized heat under high current	Adequate copper pour, thermal vias, and heatsinking in PCB design

Table 6: Project constraints and mitigations

8. Hardware and Software Requirements

8.1 Bill of Materials

Category	Item	Qty	Est. Cost (■)
Battery	Samsung INR21700-50E (3.7V, 5Ah)	6	1,800
Battery	Cell holders + nickel strips	—	300
Switching	N-MOSFET (low RDS(on), 40–80V)	18	2,700
Switching	Gate driver ICs	9	900
Sensing	INA226 current/voltage sensor (I2C)	3	450
Sensing	NTC 10K thermistors	6	120
Control	STM32F446 development board	1	800
Control	CAN transceiver module	1	150
PCB	Custom PCB fabrication	1	500–1,000
Rover	4WD/6WD chassis + DC motors	1	1,500–2,500
Rover	Motor driver (BTS7960 or L298N)	1	300–500
Misc	Connectors, wires, heatsinks	—	500
		Total	~10,000–12,000

Table 7: Estimated bill of materials

8.2 Software Tools

Purpose	Tool
PCB Design	KiCad or Altium Designer
Firmware	STM32CubeIDE / PlatformIO (C/C++)
Simulation	MATLAB/Simulink
Data Analysis	Python (matplotlib, pandas)
Communication	CAN bus (vehicle) + I2C/SPI (sensors)

Table 8: Development tools

8.3 System Architecture Summary

Layer	Components	Function
Switching Hardware	18 MOSFETs + 9 gate drivers on custom PCB	Enable series / parallel / bypass reconfiguration per cell
Sensing	3× INA226 + 6× NTC thermistors	Per-string current, per-cell voltage and temperature measurement
Control	STM32F446 MCU	Runs EKF for SOC, reconfiguration algorithm, protection logic
Communication	CAN bus + UART	Vehicle controller interface and telemetry/logging
Protection	Firmware routines	OV/UV/OC/OT/short-circuit with reconfiguration-first response

Table 9: Three-layer system architecture

9. Validation Strategy and Deliverables

9.1 Validation Methodology

Two identical rover runs will be performed: one with reconfigurable BMS active and one locked in fixed 3S2P (simulating traditional BMS). Both use the same battery pack with deliberately introduced 10–15% SOC imbalance. Metrics compared: total runtime (seconds), total energy extracted (Wh), maximum cell temperature (°C), SOC spread at end of mission (%), and number of fault events survived without mission termination.

9.2 Deliverables

Deliverable	Description
Custom PCB	MOSFET switching array with sensing, gate drivers, MCU footprint
Firmware	Reconfiguration algorithm, EKF SOC estimation, protections, CAN comm
Integrated Rover	4WD/6WD rover with reconfigurable BMS installed and operational
Simulation Model	MATLAB/Simulink model for pre-hardware algorithm validation
Test Data	Logged comparison data: reconfigurable vs fixed-topology performance
Technical Report	Complete documentation: schematics, layout, firmware, results

Table 10: Project deliverables

10. Conclusion and Future Scope

10.1 Conclusion

This project proposes a mission-adaptive reconfigurable BMS for an autonomous rover — a system that goes beyond traditional monitor-and-protect architectures by actively reconfiguring battery topology in real time. Using SOC and SOA as decision inputs, the three-switch-per-cell topology enables fault tolerance, wear leveling, adaptive current sharing, and maximum energy extraction. The system trades ~0.5W of operational overhead for 15–30% more usable energy and fault-resilient operation — a 25x return on the energy invested in BMS operation.

10.2 Future Scope

Direction	Description
Module-level scaling	Apply reconfiguration at module level for 48V+ systems (EVs, home storage)
ML integration	Augment rule-based engine with reinforcement learning for mission-specific optimization
Second-life batteries	Apply to packs with retired, heterogeneous cells where reconfiguration provides maximum benefit
Multi-platform	Port to drone and AUV platforms for cross-platform validation
Digital twin	Real-time battery pack twin for predictive degradation and long-term strategy

Table 11: Future research directions

References

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